

Taewoo Jo 조태우

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Computer Graphics and Applications Lab (CGnA), Yonsei University, Seoul, Republic of Korea

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RESEARCH INTERESTS

Extended Reality (XR) · Human-Computer Interaction (HCI) · Eye Tracking & Gaze Interaction · Accessibility · User Interface & User Experience

EDUCATION

2024 – Present	Ph.D. in Computer Science Yonsei University — Computer Graphics and Applications Lab (CGnA)
2022 – 2024	M.S. in Intelligent Robotics Gwangju Institute of Science and Technology (GIST)
2017 – 2022	B.S. in Physics Gwangju Institute of Science and Technology (GIST)
Mar – Jun 2020	Exchange Student Korea Advanced Institute of Science and Technology (KAIST)

RESEARCH EXPERIENCE

2024 – Present	Graduate Researcher Computer Graphics and Applications Lab (CGnA), Yonsei University - Bayesian modeling of implicit selection intention for manual-free gaze interaction in XR - Non-Euclidean portal reset for continuous walking in room-scale redirected walking
2020 – 2024	Research Assistant / Research Intern Human-Centered Intelligent Systems Lab (HCIS), GIST - WatchCap: wearable compensatory head-movement stimulation for people with low vision - XAI visualization for autonomous vehicles — joint research with MIT CSAIL - In-car XR platform: vehicle tracking (GPS/IMU), in-vehicle HMD tracking, driving-event-aware content - VR-based low-vision simulation and mobile wayfinding interfaces
Jul – Aug 2021	G-SURF Research Internship School of Integrated Technology, GIST

PUBLICATIONS

Bold denotes my name. * denotes equal contribution.

International Journal Papers

- [1] Ho Jung Lee, **Taewoo Jo**, Sulim Chun, In-Kwon Lee. Can't Nobody Stop Me! Non-Euclidean Portal Reset for Continuous Walking in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 32(5), 2026.

- [J2] **Taewoo Jo**, Ho Jung Lee, Sulim Chun, In-Kwon Lee. Manual-Free Gaze Interaction via Bayesian-Based Implicit Intention Prediction. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 31(12), 2025.
- [J3] Gwangbin Kim, Dohyeon Yeo, **Taewoo Jo**, Daniela Rus, Seungjun Kim. Decoding Your Ride: What Information Builds Human Trust on Real Roads? *GetMobile: Mobile Computing and Communications*, 29(2), 2025. Invited
- [J4] **Taewoo Jo**, Dohyeon Yeo, Gwangbin Kim, Seokhyun Hwang, Seungjun Kim. WatchCap: Improving Scanning Efficiency in People with Low Vision through Compensatory Head Movement Stimulation. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 8(2), Article 47, 2024.
- [J5] Hwaseung Jeon ^{*}, **Taewoo Jo** ^{*}, Dohyeon Yeo ^{*}, Eunsol An, Yumin Kang, Seungjun Kim. The Way of Water: Exploring the Role of Interaction Elements in Usability Challenges with In-Car VR Experience. *Virtual Reality*, 28(3), 2024.
- [J6] Gwangbin Kim, Dohyeon Yeo, **Taewoo Jo**, Daniela Rus, Seungjun Kim. What and When to Explain? On-Road Evaluation of Explanations in Highly Automated Vehicles. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 7(3), Article 104, 2023. Distinguished Paper
- [J7] **Taewoo Jo**, Dohyeon Yeo, Seungjun Kim. Enhancing Wayfinding Experience in Low-Vision Individuals through a Tailored Mobile Guidance Interface. *Electronics*, 12(22), 4561, 2023.

Domestic Conference Papers

- [C1] **Taewoo Jo**, Dohyeon Yeo, Seungjun Kim. A Study on User Experience of Mobile Wayfinding Application for Low Vision Pedestrians. *HCI Korea*, 2022. Best Paper

INVITED PRESENTATIONS

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|------|---|
| 2026 | Manual-Free Gaze Interaction via Bayesian-Based Implicit Intention Prediction
IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2026) — invited
TVCG journal paper presentation |
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ACADEMIC SERVICE

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| 2026 | Reviewer, ACM CHI Conference on Human Factors in Computing Systems (CHI) |
| 2026 | Reviewer, IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) |
| 2026 | Reviewer, ACM Symposium on User Interface Software and Technology (UIST) |
| Nov 2025 | Reviewer, Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT) |

AWARDS, HONORS AND SCHOLARSHIPS

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| 2024 – Present | BK21 FOUR Scholarship (Full Funding), Yonsei University |
| 2023 | Distinguished Paper Award, ACM IMWUT / UbiComp 2023 |
| 2022 – 2024 | GIST Scholarship (M.S., Government Supported) |
| 2022 | Best Paper Award, HCI Korea 2022 |
| 2017 – 2022 | GIST Scholarship (B.S., Government Supported) |

WORK EXPERIENCE

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| Sep – Dec 2022 | Metaverse Fellowship
RAPA × Meta |
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Oct 2022	Pitch Your Lab GIST Metaverse Research Center, ISMAR 2022
Apr – Jun 2021	External Developer Unity Simulation Development, Robotry Co.
Dec 2019 – Feb 2020	Industrial-Academic Internship CUop, CREATY design & Breez design
2019 – Present	Senior Mentee Hanlim-Mirae Science Camp, The Korean Academy of Science and Technology (KAST)

TEACHING EXPERIENCE

Spring 2024	Computer Graphics — Teaching Assistant, Yonsei University
Spring 2023	Human-Computer Interaction — Teaching Assistant, GIST
2021 – 2022	Mathematics in Civilization — Teaching Assistant, GIST
Spring 2021	General Physics — Teaching Assistant, GIST
Fall 2020	Introduction to Linear Algebra — Teaching Assistant, GIST

TECHNICAL SKILLS

Languages	C#, C++, Python
Development	Unity 3D, XR / VR (Meta Quest Pro, HMD tracking), eye tracking
Analysis	SPSS, JASP
Modeling	Rhino, Autodesk 360